

## UNIT-1: Security Concepts (The Foundation)

*Target: Full Marks (Easy Theory)*

1. **Network Security Model:** Define the model for network security with a neat block diagram (Sender, Receiver, Opponent, Trusted Third Party). Explain the role of each component.
2. **Attacks, Services, Mechanisms:**
  - Define **Security Attacks** and distinguish between **Passive vs. Active attacks**.
  - Explain the **Security Services** (Confidentiality, Integrity, etc.) and **Security Mechanisms** defined in X.800.
3. **Classical Encryption:**
  - Explain **Playfair Cipher** and **Transposition Techniques** with examples.
  - Differentiate between **Substitution** and **Transposition** ciphers.
4. **Steganography:** What is Steganography? How is it different from Cryptography?

- Differentiate between **Passive** and **Active** attacks with examples.
- Explain the **Security Services** (Confidentiality, Authentication, Integrity, Non-repudiation, Access Control) defined by X.800.

### 3. **Classical Algorithms:**

- Explain **Playfair Cipher** with an example matrix.
- Explain **Transposition Ciphers** (Row Transposition / Rail Fence) with examples.

### 4. **Steganography:** Compare and contrast **Steganography** and **Cryptography**. What are the advantages and disadvantages of Steganography?

### **Short Answer (2-3 Marks):**

- Define Security Mechanism.
- What is the difference between Interruption and Interception?
- Define Non-Repudiation.

## UNIT-II: Symmetric & Asymmetric Ciphers

*Goal: Memorize the diagrams and the RSA steps.*

### Long Answer Questions (10 Marks):

1. **DES:** Draw the general block diagram of **DES** and explain the **Feistel Cipher structure** in detail. (Do not memorize the S-Box numbers, just the logic).
2. **AES:** Explain the **AES** encryption process. Describe the four transformations: **Substitute Bytes, Shift Rows, Mix Columns, and Add Round Key**. (★ ★ ★ Gunshot)
3. **Block Modes:** Explain the Block Cipher Modes of Operation: **ECB, CBC, CFB, and OFB** with block diagrams.
4. **RSA Algorithm:** Explain the RSA algorithm for encryption and decryption.
  - *Problem:* Perform encryption and decryption using RSA for  $p = 3, q = 11, e =$

block diagrams.

4. **RSA Algorithm:** Explain the RSA algorithm for encryption and decryption.

- *Problem:* Perform encryption and decryption using RSA for  $p = 3, q = 11, e = 7, M = 5$ .

5. **Diffie-Hellman:** Explain the **Diffie-Hellman Key Exchange** protocol. How is it vulnerable to Man-in-the-Middle attack?

**Short Answer (2-3 Marks):**

- What is the difference between Block Cipher and Stream Cipher?
  - Define the "Avalanche Effect".
  - What are the key sizes for DES and AES?
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## UNIT-II: Symmetric & Asymmetric Ciphers (The Core)

*Target: Diagrams & Algorithms. Note the specific algorithms mentioned.*

### 1. Block Cipher Principles & Modes:

- Explain the **Block Cipher Modes of Operation** (ECB, CBC, CFB, OFB) with diagrams.
- Explain the **Feistel Cipher Structure** (used in DES).

### 2. Symmetric Algorithms (DES & AES):

- Explain the **AES (Advanced Encryption Standard)** structure. Detail the 4 transformations: *Substitute Bytes, Shift Rows, Mix Columns, Add Round Key*.
- Explain the general structure of **DES**.

### 3. Asymmetric Algorithms (RSA & DH):

- **RSA:** Explain the RSA algorithm (Key Generation, Encryption, Decryption). **Practice a**

- Explain the general structure of **DES**.

### 3. **Asymmetric Algorithms (RSA & DH):**

- **RSA:** Explain the RSA algorithm (Key Generation, Encryption, Decryption). **Practice a problem** (e.g.,  $p = 3, q = 11$ ).
- **Diffie-Hellman:** Explain the Diffie-Hellman Key Exchange algorithm.

### 4. **Minor Algorithms (Short Notes):**

- Write a short note on **RC4** (Stream Cipher) or **Blowfish**.
- Explain the **Knapsack Algorithm** (This is explicitly mentioned in R22 Asymmetric Ciphers).

## **UNIT-III: Hash Functions & Key Management**

*Target: Logic Diagrams.*

### **1. Hash Functions:**

- Explain the **SHA-512** logic and message processing. (Draw the block diagram showing the 1024-bit block).

### **2. Message Authentication:**

- Explain **HMAC** design objectives and structure.
- Explain the **Elgamal Digital Signature Scheme**.

### **3. Key Management:**

- Explain **X.509 Authentication Service** and Certificate format.
- Explain **Kerberos** (how it works for authentication).

3. **Digital Signatures:** Explain the ElGamal Digital Signature scheme. How is it different from RSA?

4. **Key Management:**

- Explain the **X.509 Authentication Service** and the format of the X.509 Certificate.
- Explain the motivation and working of **Kerberos** (Version 4).

**Short Answer (2-3 Marks):**

- What are the requirements of a Hash Function?
- What is a Digital Signature?
- Differentiate between Weak Collision Resistance and Strong Collision Resistance.



## **UNIT-V: Email, IP Security & Case Studies**

*Target: The R22 Specific Questions are here.*

### **1. E-Mail Security:**

- Explain **PGP (Pretty Good Privacy)** services and message format.

### **2. IP Security (IPSec):**

- Explain **IPSec Architecture** and **Security Associations (SA)**.
- Differentiate between **Transport Mode** and **Tunnel Mode**.